

Web Project

Learning Management System

Software Requirements

Team Members

ID	Name	Email
21038519	Yara Hossam Elden Wahed	jarahossam97591@gmail.com
21047496	Hania Hisham Ahmed	haniahisham4@gmail.com
21060177	Nada Hossam Ahmed	nadahossamawad@gmail.com
21047445	Nada Ahmed Ezzat	nadaahmed0100@gmail.com
21039956	Abdelrahman Ahmed Abdelrahman	abdelrahmanahmed6335@gmail.com
21058818	Ali Mohsen Abd El Monsef	ali.m.abdelmonsef@gmail.com

Group Code

CLS_GIZ3_SW5D5_S1

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Document Purpose and Audience

Document Purpose

- The purpose of this document is to define the requirements and specifications for the Learning Management System (LMS). It provides a detailed description of the

system's features, functionalities, and constraints to ensure that all stakeholders have a common understanding of the project objectives.

Intended Audience

- This document is intended for the following stakeholders:
- **Administrators (System Admins)** – To understand management-related features and access control mechanisms.
- **Instructors** – To gain clarity on the tools available for course creation, delivery, and assessment.
- **Students (End Users)** – To be aware of the features available for learning, progress tracking, and certification.
- **Guest** – To browse and search courses with limited access and decide whether to register for full features.

Introduction

Software Purpose

The Learning Management System (LMS) is a web-based platform designed to facilitate online education by connecting **students, instructors, and administrators** in a structured environment. The system provides features such as course management, user enrollment, online assessments, performance tracking, notifications, and certification. It ensures a secure, scalable, and user-friendly solution for managing the complete learning lifecycle.

Objectives

- Provide an accessible platform for students to enroll in courses, access learning materials, and track their progress.
- Enable instructors to create and manage courses, assignments, and assessments efficiently.
- Offer administrators full control over user management, course approvals, and overall system monitoring.
- Enhance the learning experience with interactive tools such as quizzes, live lectures, notifications, and automated certificates.

By centralizing these functionalities, the LMS aims to **improve learning outcomes, streamline course management, and enhance communication** between all stakeholders.

Software Scope

The LMS will include the following functionalities:

1. **User Management:** Registration, authentication, profile management, and role-based access (Admin, Instructor, Student, Guest).
2. **Course Management:** Course creation, enrollment, lecture uploads, assignments, quizzes, and grading.
3. **Assessment & Grading:** Online quizzes, assignment submissions, and feedback.
4. **Notification System:** Alerts for deadlines, results, enrollments, and reminders via in-system and email notifications.
5. **Certificate Management:** Auto-generated completion certificates in PDF format.
6. **Security & Access Control:** Role-based permissions, secure authentication, data privacy, and session management.
7. **Non-Functional Features:** Scalability, high performance, usability, mobile responsiveness, and integration with third-party tools (Zoom, Google Meet, payment gateways).

Project Plan

- **Timeline**

Week	Tasks / Activities	Milestones	Deliverables	Resources
Week 1	- Define project requirements - Initial research Draft project scope & objectives	Project Kick-off	Project proposal & scope document	Project Manager, Analyst
Week 2	- Design wireframes & user journey - Database schema design - UI/UX initial design	Design Approval	Wireframes + UI draft	Designer, Developer
Week 3	- Start core development (backend + frontend setup) - API integration (if needed) - Internal testing (basic)	Core System Ready	Functional prototype	Developers, QA
Week 4	- Final testing & bug fixing - Documentation (user manual, tech doc) - Final presentation/demo	Project Closure	Final product + documentation	Full Team

Milestones

- M1 (End of Week 1): Requirements finalized & System design completed (ERD, UML diagrams, UI/UX wireframes).
- M2 (End of Week 2): Database ready + core LMS features implemented.
- M3 (End of Week 3): Core LMS features implemented (user management, course management, basic assessments).
- M4 (End of Week 4): Testing completed (>80% pass rate) + Final submission & presentation.

Deliverables

1. UML Diagrams (Use case, Class, Activity, Sequence, ERD)
2. Database Schema (Logical & Physical)
3. UI/UX Wireframes
4. Source Code + README
5. Test Plan & Bug Report
6. Final Report + Presentation

Resource Allocation by Week

- **Week 1–2 (Requirements & Design)**
 - PM → Finalize requirements, risk assessment.
 - Backend Dev → Draft ERD + database schema.
 - Frontend/UI → Create wireframes & mockups.
 - Documentation Lead → Project proposal + requirements doc.
 - Tester → Prepare initial test plan.

- **Week 3 (Implementation – Core LMS Features)**

- Backend Dev → Implement user management + course management modules.
- Frontend/UI → Build UI pages (login, dashboard, course view).
- PM → Monitor progress, resolve blockers.
- Tester → Start unit testing.
- Documentation Lead → Update system design doc.

- **Week 4 (Testing & Final Delivery)**

- Backend Dev → Fix bugs, finalize features.
- Frontend/UI → Polish UI + integrate backend.
- Tester → Run test cases, document results.
- PM → Ensure milestones met, prepare for presentation.
- Documentation Lead → Compile final report + PPT.

- **Tools & Technologies**

- Development: Java/Spring Boot, MySQL, HTML/CSS/JS.
- Design: Figma (UI/UX), Lucidchart/Draw.io (ERD/UML).
- Collaboration: GitHub (version control), Google Drive (docs).
- Testing: JUnit, manual test cases.

Requirements

User Management

- **User Types**

- **Admin** (Super user, only one exists in the system)
- **Instructor**
- **Student**
- **Guest**

- **Admin Capabilities**

- Approve or reject user registration applications.
- Create, update, and delete user accounts.
- Update personal admin account.
- Manage system-level entities (skills, categories, etc.).

- **Instructor Capabilities**

- Cannot manage users.
- Request enrollment as an instructor in specific courses.
- Manage course-related content (assignments, tests, lectures).

- **Student Capabilities**

- Cannot manage users.
- Focused only on enrolling, learning, and interacting with course content.

- **Guest Capabilities**

- Can only browse and search courses.
- Must register to access advanced features.

- **Features**

- **User Registration & Login** (with role-based access control).
- **Profile Management** (view/update personal information).

- Password recovery, email verification.
- Account activation/deactivation by admin.

Course Management

Admin Capabilities

- **CRUD** (Create, Read, Update, Delete) courses.
- Approve user enrollment in courses (instructors Only).
- Remove users from courses.
- Manage **categories & skills** (only admin can).
- Track course progress (all users).
- Upload recorded lectures (if needed).

Instructor Capabilities

- Request enrollment as instructor.
- Browse courses.
- Create & manage **assignments and quizzes/tests**.
- Launch **live lectures/meetings**.
- Upload recorded lectures.
- Upload lecture materials (slides, PDFs, docs).
- Grade assignments and quizzes.

Student Capabilities

- Enroll in courses (with admin approval if required).
- Browse courses.
- View course details.
- Watch Recorded lecture videos.
- Join Live Lectures.
- Download lecture materials.
- Take quizzes/tests.
- Submit assignments.
- Track personal progress.
- View grades.
- Download certificates of completion.
- Rate and review courses.

Guest Capabilities

- Browse courses.
- Search courses by category, skill, or keyword.
- Register to gain more access.

3. Assessment & Grading

Assessment Types

- **Quizzes** (MCQ, T/F, short answers)
- **Assignments** (File upload / text submission)

Quiz Management

- Instructors can create quizzes per course.
- Questions bank maintained per course.
- Randomized questions for each student attempt.
- Auto-grading for MCQ & T/F.
- Manual grading for short answers.

Assignment Management

- Students can upload files as submissions.
- Instructors review & grade manually.
- Plagiarism check (optional future feature).

Feedback

- **Automated Feedback** (for quizzes – grades, wrong answers, correct answers).
- **Manual Feedback** (for assignments – instructor comments, grade).

4. Performance Tracking

- **Student Progress Tracking:**
 - a. Scores (quizzes, assignments).
 - b. Attendance in live lectures.
 - c. Course completion status.

- **Instructor Tracking:**
 - a. Student performance reports.
 - b. Participation statistics.
- **Admin Tracking:**
 - a. Overall course performance.
 - b. Instructor activity reports.

5. Notification System

- **For Students:**
 - a. Enrollment confirmation.
 - b. Assignment grading results.
 - c. Quiz results (automated).
 - d. Upcoming deadlines/reminders.
- **For Instructors:**
 - a. New student enrollments.
 - b. Assignment submissions waiting for grading.
 - c. Course-related system alerts.
- **Features:**
 - a. Separate “Unread” vs “All” notifications.
 - b. Email notifications in addition to in-system alerts.

6. Certificate Management

- Certificates generated automatically after course completion.
- Includes student name, course name, completion date, and instructor/admin signature.
- Downloadable in PDF format.
- Stored in student’s profile for future access.

7. Course Review & Rating

- Students can rate courses (e.g., 1–5 stars).
- Students can submit written reviews.
- Instructors and Admins can view course feedback.
- Admin moderates inappropriate reviews.

8. Security & Access Control

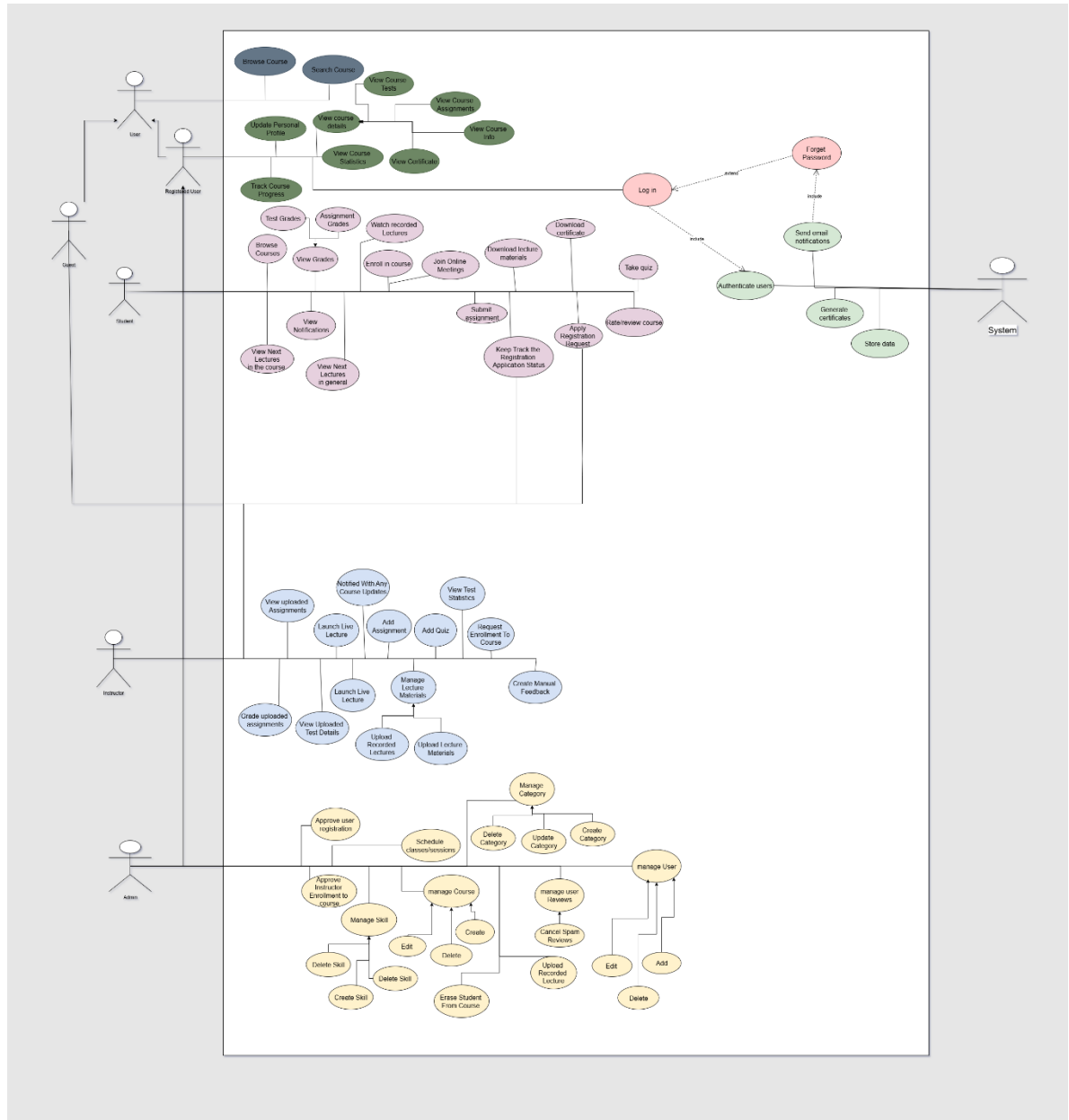
- **Authentication:**
 - a. Implemented via **Spring Security** (if using Java/Spring).
 - b. Passwords hashed & stored securely.
- **Authorization:**
 - a. Role-based access control (RBAC).
 - b. Permissions based strictly on user type (Admin, Instructor, Student, Guest).
- **Session Management:**
 - a. Secure login sessions.
 - b. Automatic logout after inactivity.
- **Data Privacy:**
 - a. User information secured and GDPR-compliant (if required).

Non Functional Requirements

Requirement	Details
Performance	• Should support hundreds/thousands of concurrent users. • Optimized DB queries for fast search & reporting.
Scalability	• Should allow easy integration of new roles/features. • Support for cloud storage (lectures, materials, backups).
Usability	• Mobile-friendly (responsive UI). • Easy navigation, accessibility (WCAG compliance).
Reliability	• Regular data backup and restore mechanisms. • 99.9% uptime with hosting/cloud solutions.
Integration	• Support for integration with video conferencing tools (Zoom, MS Teams, Google Meet). • Payment gateway integration (if monetized).

System Models

Use Case Model



Use Case Tables

Use Case ID:		
Use Case Name:	Create Assignment	
Actors:	Instructor	
Pre-conditions:	The instructor should be logged in. The Email should have the instructor role.	
Post-conditions:	The assignment is uploaded and viewable by students.	
Flow of events:	User Action	System Action
	1- The instructor goes to create assignment page	
		2- The system verifies authorization. and show the create page
	3- The instructor chooses the assignment file and uploads the file.	
		4- system shows that the file uploaded and show the ability to remove the uploaded file
	5- The instructor presses the submit button.	
		6- The system shows that the assignment file is submitted.
Exceptions:	User Action	System Action
	1- The instructor chooses the file.	

		2- The file is not with the PDF extension.
		3- The system rejects the submission and shows the user the reason.
Includes:	No use cases included.	

Use Case ID:		
Use Case Name:	Create Quiz	
Actors:	Instructor	
Pre-conditions:	The instructor should be logged in. The email should have the instructor role.	
Post-conditions:	The quiz is uploaded and viewable by students.	
Flow of events:	User Action	System Action
	1- The instructor goes to create quiz page	
		2- The system verifies authorization. and show the create page
	3- The instructor enters the number of questions that should be in the quiz, the duration of the quiz, and the start date and time of the quiz.	
	4- The instructor creates a quiz.	
		5- The system chooses random questions from the question bank of the course.
		6- The system shows that the quiz is ready and will start on Date.

Exceptions:	User Action	System Action
	1- The instructor chooses the file.	
		2- The file is not with the PDF extension.
		3- The system rejects the submission and shows the user the reason.
Includes:	No use cases included.	

Use Case ID:		
Use Case Name:	Register	
Actors:	Guest User (unregistered)	
Pre-conditions:	The user has not registered before or is not logged in	
Post-conditions:	A user account is created in the system	
Flow of events:	User Action	System Action
	1-Guest user clicks on "Register" or "Sign Up" button	
		2- System displays registration form with required fields
	3- Guest fills in registration details	
		4- System validates input fields
	5- Guest clicks "Submit" or "Create Account" button	
		7- The system checks if the email already exists in the database.
		8- system creates new user account

		9- System displays pending confirmation account message and send application to admin dashboard
	10-admin approve/decline user application	
		11-system sends notification to user
Exceptions:	User Action	System Action
	1- User enters invalid email format	
		2- system displays an error message.
	3- User enters password that doesn't meet requirements	
		4- System displays error message "Password must be at least 8 characters with uppercase, lowercase, and numbers."
	5- User enters email that already exists	
		6- System displays error message "This email is already registered. Please log in or use forgot password"
	7- User leaves required fields empty	
		8- System displays error message "Please fill in all required fields."
	9- User's password and confirm password don't match	
		10- System displays error message "Passwords do not match"

Includes:	- Email validation - Password encryption - Send confirmation email
Notes and Issues:	

Use Case ID:		
Use Case Name:	student joins the course	
Actors:	Student	
Pre-conditions:	Students should be logged in. The email should have the student role.	
Post-conditions:	Student join to course	
Flow of events:	User Action	System Action
	1- Student requests course page	
		2- The system shows the page that contain all of the active courses
	3- student write string to search courses	
		4- System filters courses based on the word
	5- The student chooses a course to get its details.	
		6- system show the course details
	7- Student applies to course	
		8- System verification of overlapping
	9- Student joins the course	

Exceptions:	User Action	System Action
	1- Student requests course details	
		2- system shows course details
	3- The student requests to join the course.	
		4- The system finds out that the student has overlapping courses.
		5- The system rejects the user request.
Includes:	Log in	

Use Case ID:		
Use Case Name:	Instructor applies for courses	
Actors:	Instructor, Admin	
Pre-conditions:	Instructors should be logged in. The email should have the instructor role.	
Post-conditions:	Instructors join the course	
Flow of events:	User Action	System Action
	1- Instructors get courses page	
		2- The system shows the courses page
	3- Instructors write strings to search courses.	
		4- System filters courses based on the word.
	5- Instructors choose a course to get its details.	

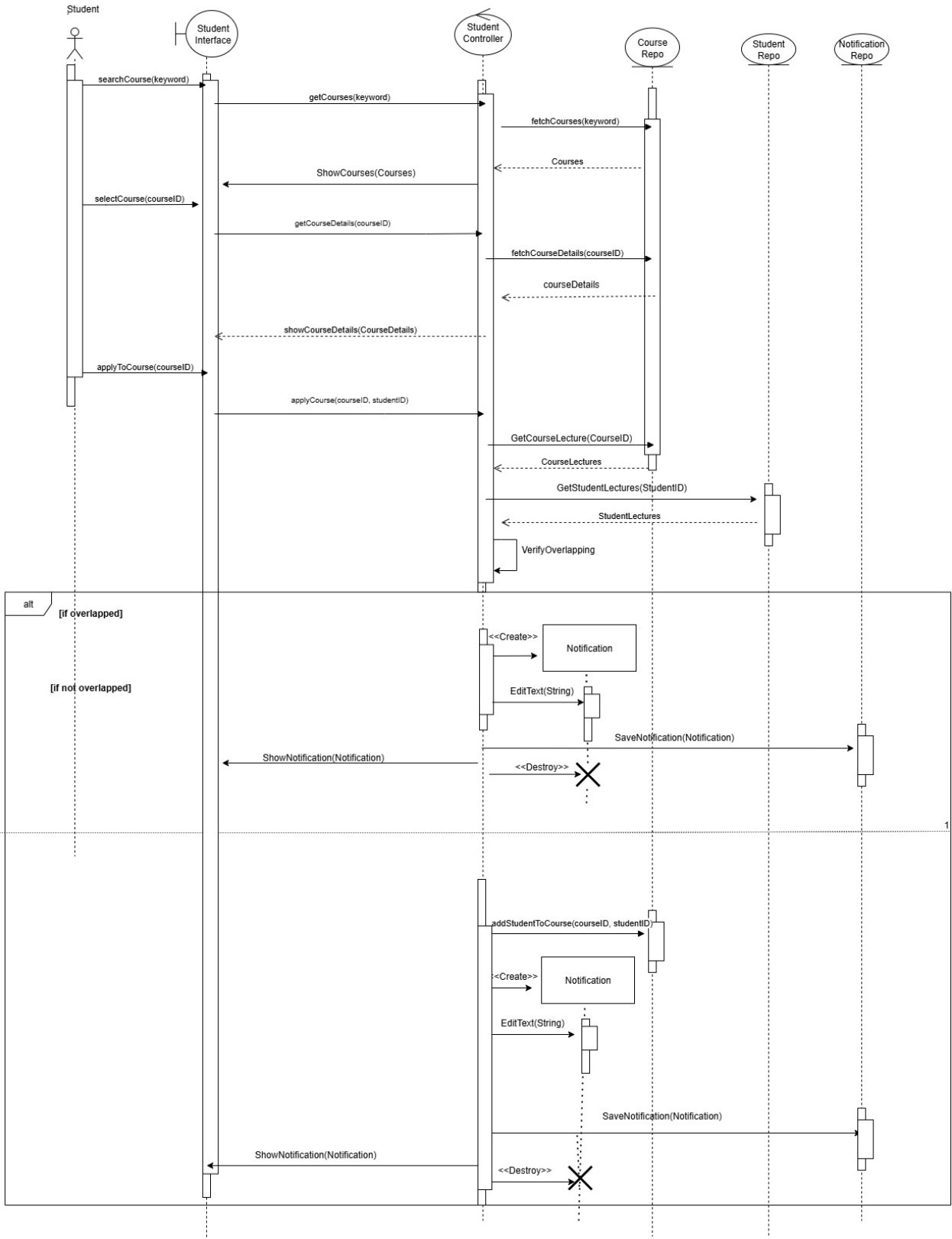
		6- system show the course details
	7- Instructor applies to course	
		8- System verification of overlapping
		9- The system sends the application to the admin.
	10- Admin request applications page	
		11- System Show Applications Page
	12- Admin request to see application details.	
		13- The system shows application details.
	14- Admin Approve application	
		16- System Send Notification to Instructor
Exceptions:	User Action	System Action
	1- Admin View Application Details	
		2- system shows Application details
	3- Admin Deny Application	
		4- System Send Notification to Instructor
Includes:	Log in	

Use Case ID:		
Use Case Name:	Lecture Establishment	
Actors:	Instructor	
Pre-conditions:	Instructors should be logged in. The email should have the instructor role.	
Post-conditions:	Instructors join to course	
Flow of events:	User Action	System Action
	1- Instructor goes to the lecture page.	
		2- The system shows the Lecture Page
	3- instructor creates Zoom Meeting and uploads the invitation link	
		4- system sends to all students join to this course an email a lecture has started
Includes:	Log in	

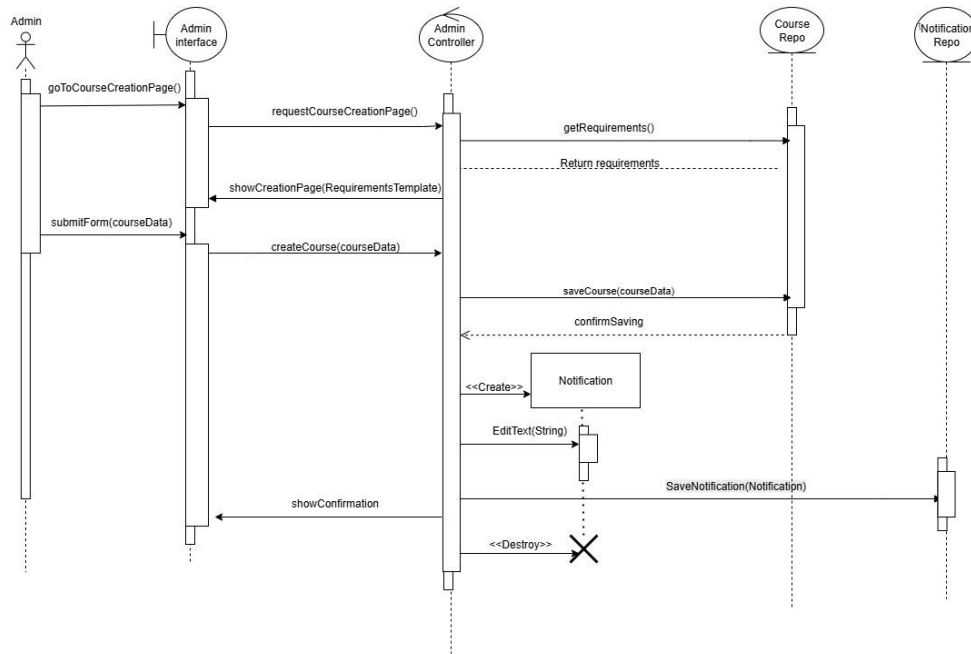
Use Case ID:		
Use Case Name:	Course Creation	
Actors:	Admin	
Pre-conditions:	Admin should be logged in. The email should have the admin role.	
Post-conditions:	Course Created successfully	
Flow of events:	User Action	System Action
	1- Admin goes to the course creation page.	

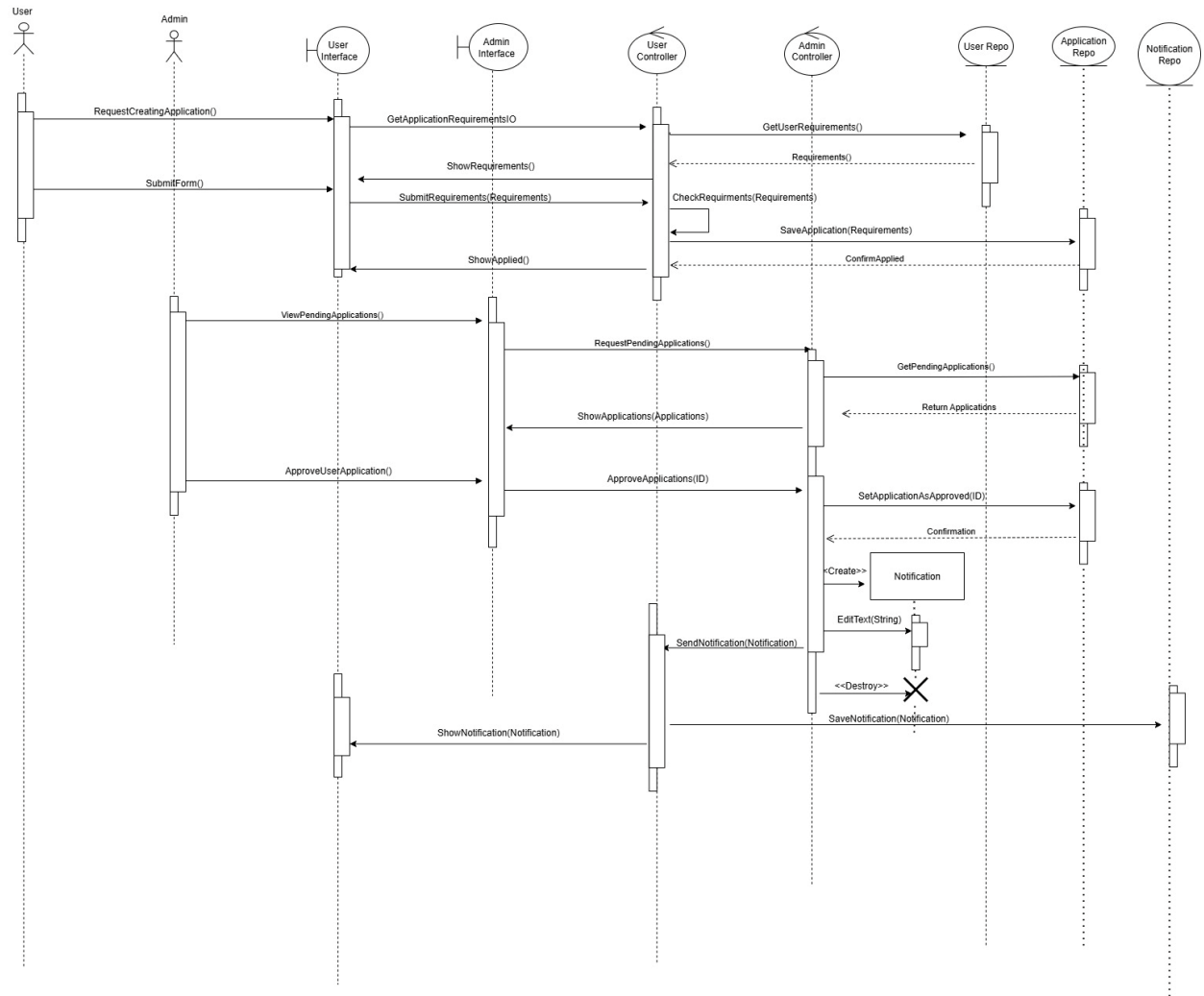
		2- The system shows the Course Creation Page
	3- Admin fills requirements and press submit	
		4- system check requirements and add the course to courses
Exceptions:	User Action	System Action
	1- Admin upload course requirement	
		2- system check requirements
		3- system finds out that requirements are in wrong format
		5- The system rejects the admin request and shows an error message.
Includes:	Log in	

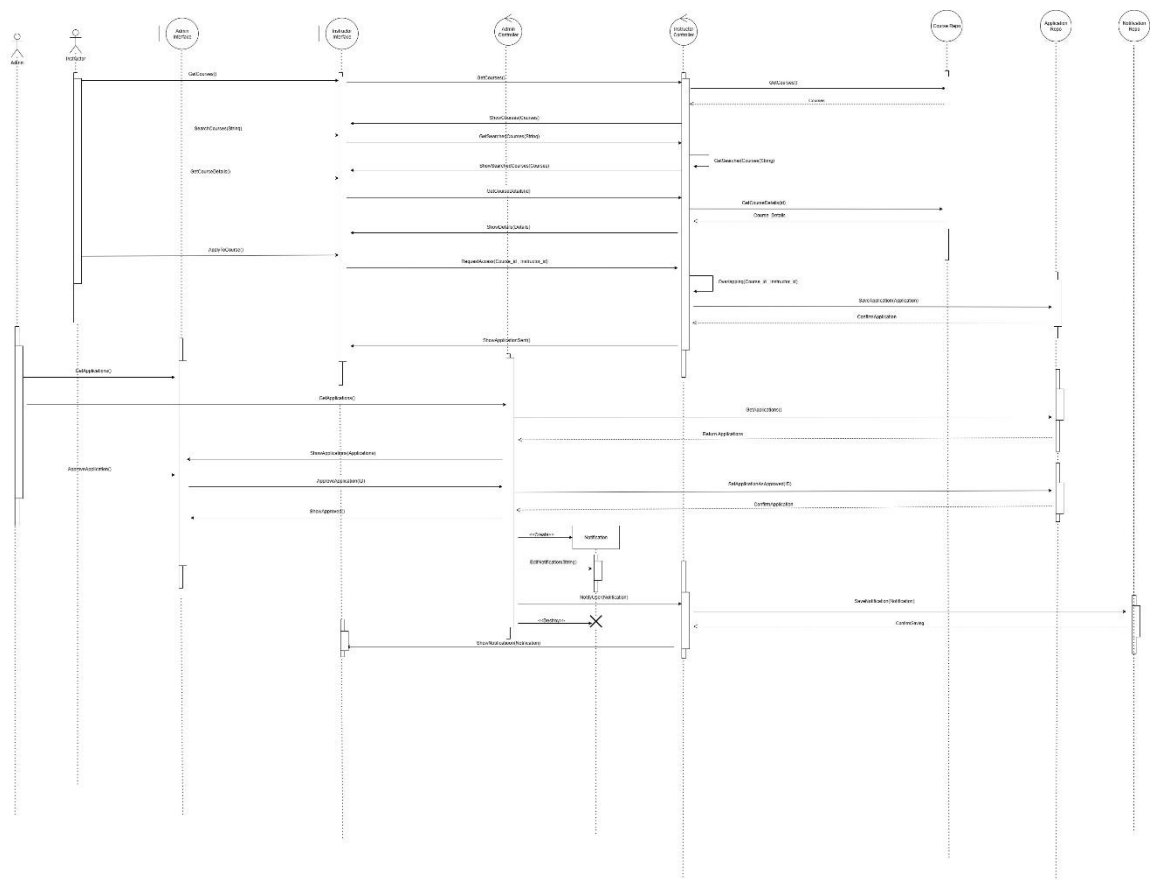
Sequence Diagrams

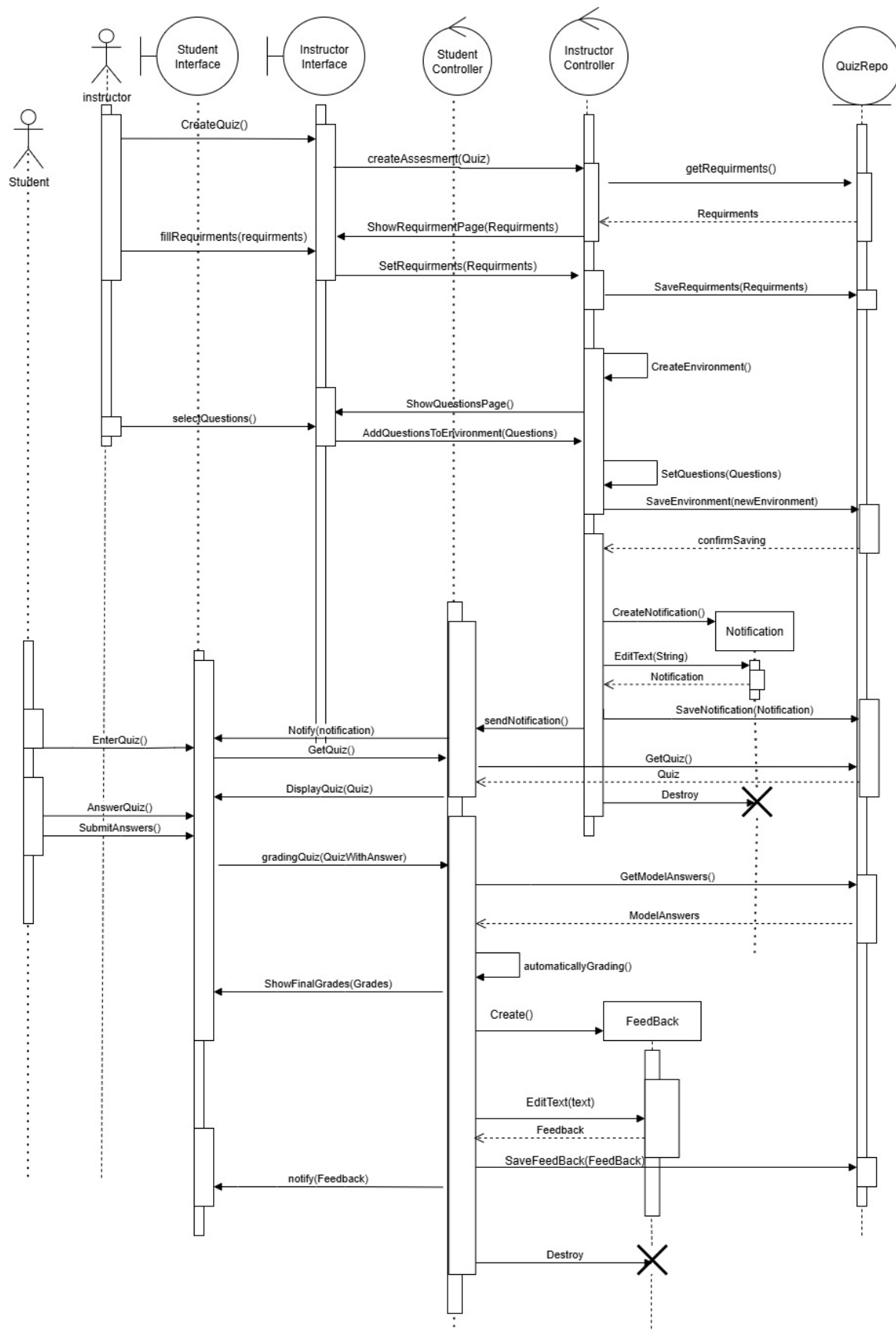


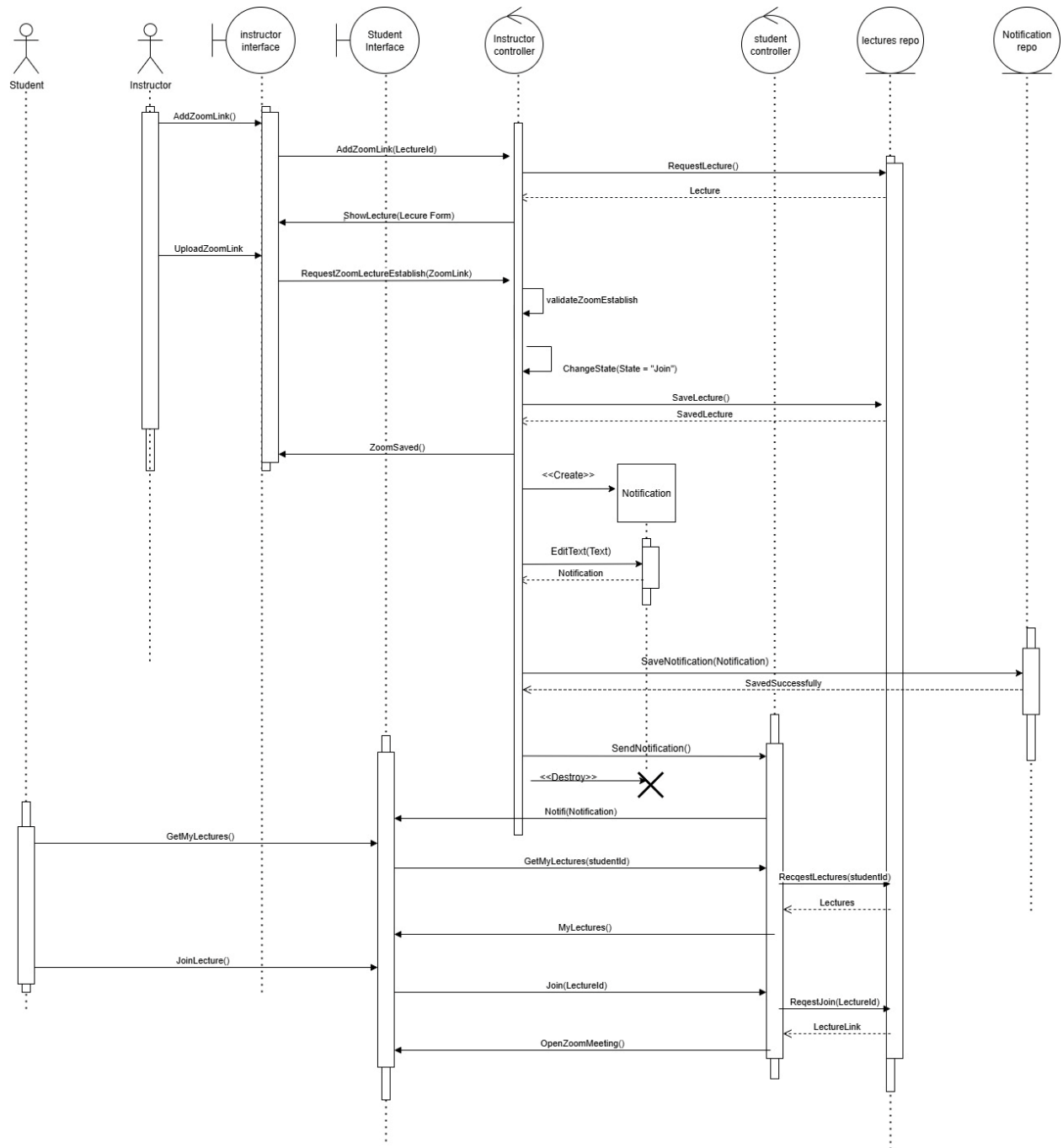
Create Course

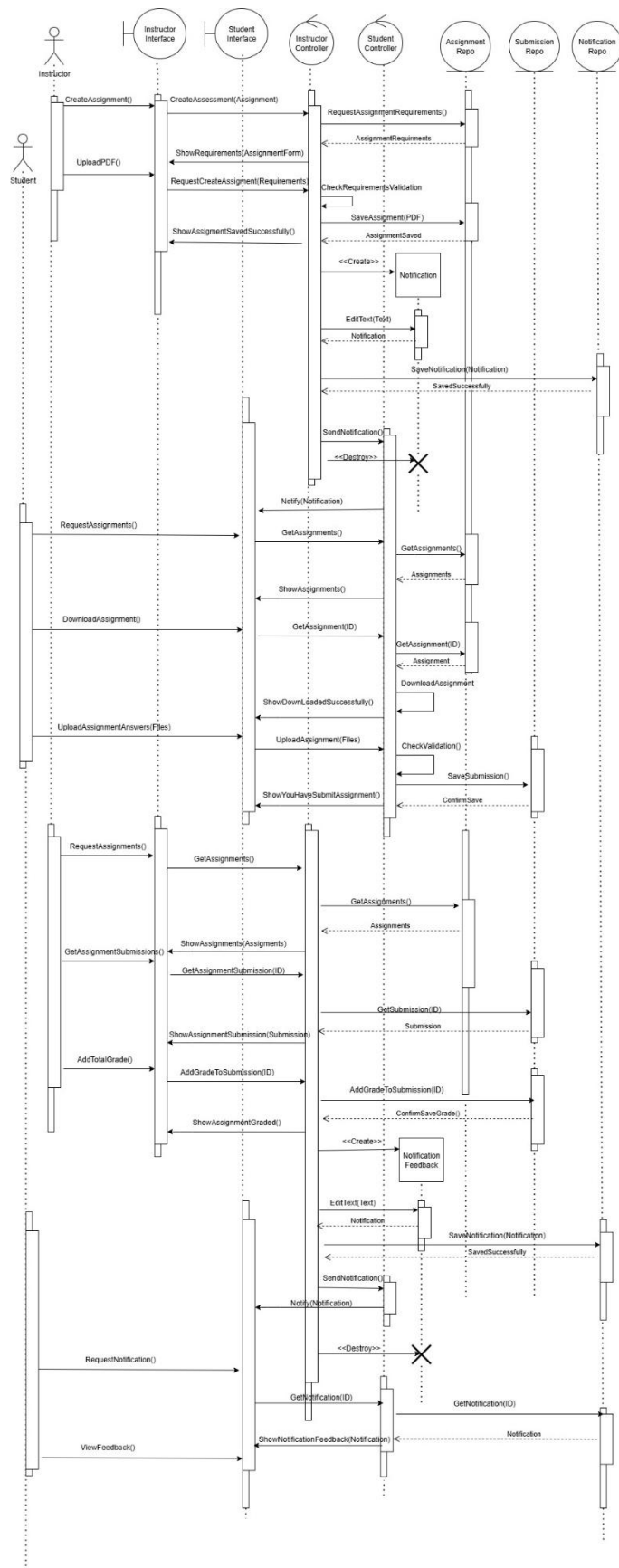




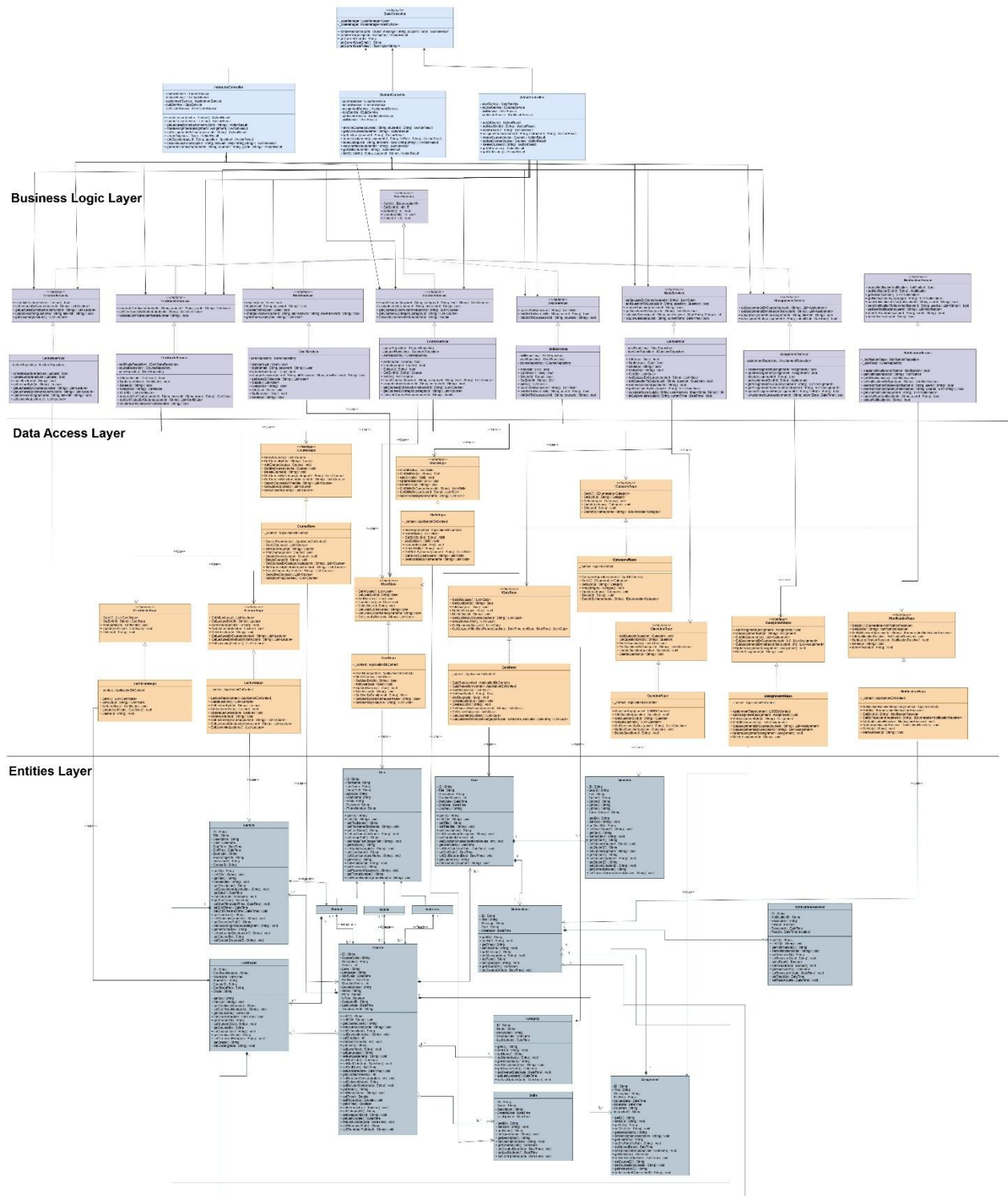








Presentation Layer



ERD

